**2** Figure 2.1 shows a Kundt's tube of length 35 cm, containing air and fine sand, having one end open at position **M** and one closed at position **N**. A loudspeaker connected to a sound generator is placed at its open end. When the sound generator and loudspeaker is turned on, four small heaps of sand eventually formed at certain positions in the tube. Adjacent heaps of sand are at distance 10 cm apart as shown in Figure 2.1.

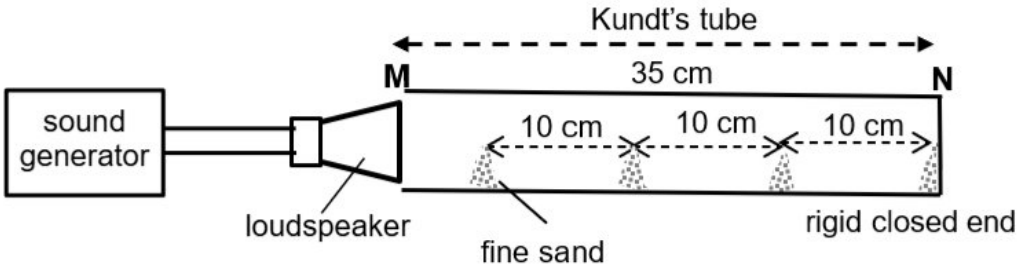



**Figure 2.1**

**(a)** Explain how a stationary wave is formed in the tube.

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..... [2]

**(b)** Determine the wavelength of the sound produced by the loudspeaker.

wavelength = ..... cm [1]

**(c) (i)** The sound wave produced by the loudspeaker in Figure 2.1 has amplitude of $A_o$.

On Figure 2.2, sketch a graph of the variation of amplitude, in terms of $A_o$, of the stationary sound wave with distance $x$ along the Kundt's tube, starting from point **M** of the Kundt's tube ($x$ = 0 cm), until its closed end at point **N** ($x$ = 35 cm). You may assume there is no energy loss by the sound wave as it is reflected at the closed end of the tube.

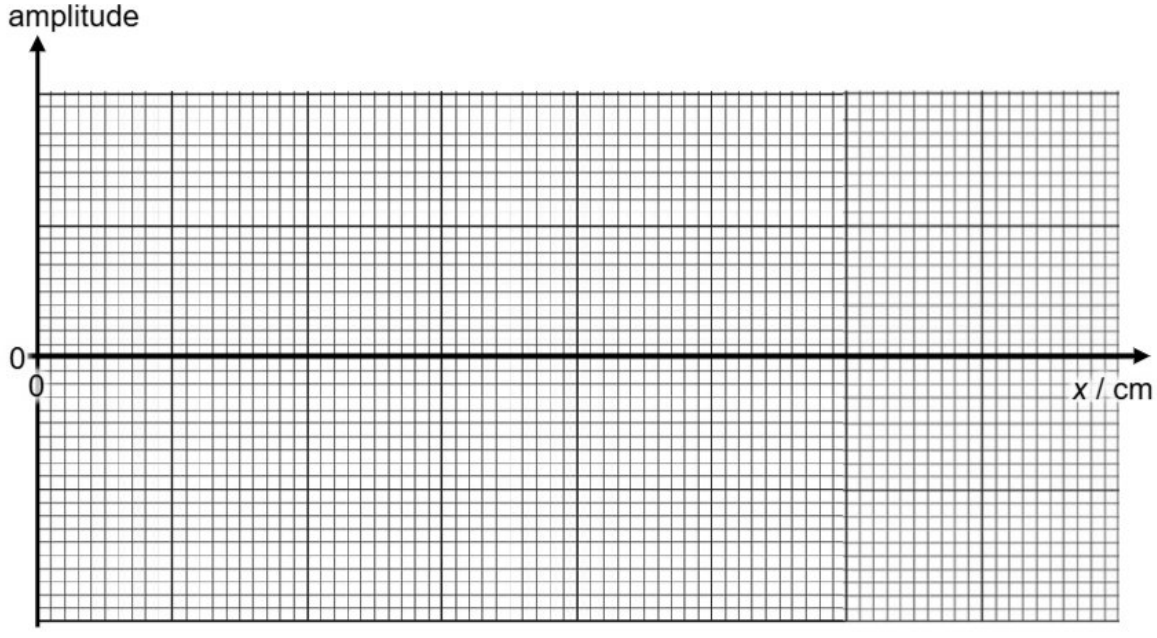



**Figure 2.2** [3]

**(ii)** By describing the movement of air molecules in the stationary sound wave, state and explain where the air pressure varies the most.

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..... [3]

**(iii)** The maximum air pressure in the Kundt's tube is $P_o$.

On Figure 2.3, sketch a graph of air pressure variation, in terms of $P_o$, against distance $x$ along the Kundt's tube from point **M** ($x$ = 0) to point **N** ($x$ = 35 cm).

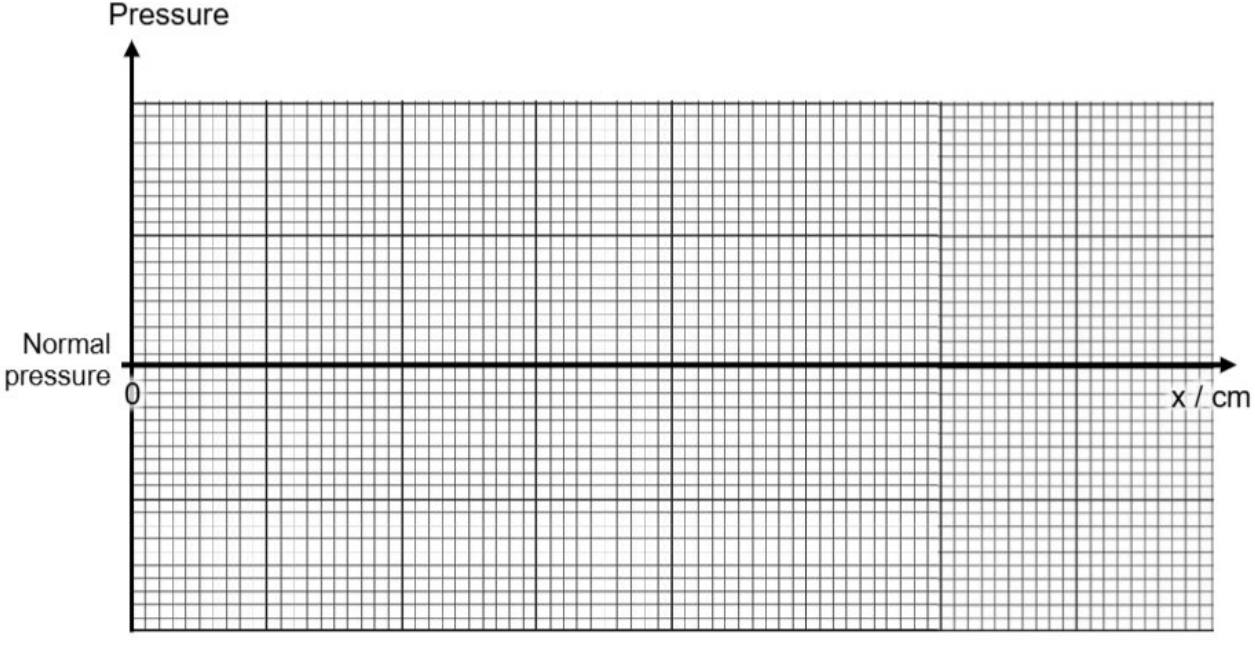



**Figure 2.3**

[2]

[Total: 11]